Sums and products of zeroes of polynomials — Solutions

Warm-up

Question 1:

$$\alpha+\beta=-\frac{-5}{1}=5$$
$$\alpha\beta=\frac{6}{1}=6$$

Question 2:

$$\alpha+\beta+\gamma=-\frac{4}{1}=-4$$

Question 3:

$$\alpha\beta+\beta\gamma+\gamma\alpha=\frac{4}{2}=2$$

Question 4:

$$\alpha\beta\gamma=-\frac{-5}{3}=\frac{5}{3}$$

Question 5:

$$\begin{aligned}(x-2)(x-3)(x+1)&=0\\(x^2-5x+6)(x+1)&=0\\x^3+x^2-5x^2-5x+6x+6&=0\\x^3-4x^2+x+6&=0\end{aligned}$$

Question 6:

$$\begin{aligned}\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}&=\frac{\beta\gamma+\alpha\gamma+\alpha\beta}{\alpha\beta\gamma}\\&=\frac{2}{5}\end{aligned}$$

Question 7:

$$0^2=\alpha^2+\beta^2+\gamma^2+2(-7)$$
$$0=\alpha^2+\beta^2+\gamma^2-14$$
$$\alpha^2+\beta^2+\gamma^2=14$$

Question 8:

$$\alpha+\beta+\gamma=-\frac{0}{4}=0$$

Question 9:

$$4\gamma=8\implies\gamma=2$$
$$P(2)=2^3-2(2^2)+k(2)-8=0$$
$$8-8+2k-8=0$$
$$2k=8\implies k=4$$

Question 10:

$$\begin{aligned}(\alpha+1)(\beta+1)(\gamma+1)&=\alpha\beta\gamma+(\alpha\beta+\beta\gamma+\gamma\alpha)+(\alpha+\beta+\gamma)+1\\&=r+q+p+1\end{aligned}$$

Skill-building

Question 1:

$$\begin{aligned}\alpha^2+\beta^2+\gamma^2 &= (\alpha+\beta+\gamma)^2 - 2(\alpha\beta+\beta\gamma+\gamma\alpha)\\ &= 4^2 - 2\left(\frac{5}{2}\right)\\ &= 16-5=11\end{aligned}$$

Question 2:

$$\begin{aligned}\alpha^2\beta\gamma+\alpha\beta^2\gamma+\alpha\beta\gamma^2 &= \alpha\beta\gamma(\alpha+\beta+\gamma)\\ &= (-r)(-p)=pr\end{aligned}$$

Question 3:

$$\begin{aligned}(\alpha+\beta)(\beta+\gamma)(\gamma+\alpha) &= (-2-\gamma)(-2-\alpha)(-2-\beta)\\ &= -(2+\alpha)(2+\beta)(2+\gamma)\\ &= -P(-2)\\ &= -((-2)^3+2(-2)^2+3(-2)+4)\\ &= -(-8+8-6+4)=-(-2)=-2\end{aligned}$$

Question 4:

$$\begin{aligned}\alpha\beta\gamma=4 &\implies \gamma^2=4 \implies \gamma=2\\ \alpha+\beta+\gamma=6 &\implies \alpha+\beta=4\\ a&=\alpha\beta+\gamma(\alpha+\beta)\\ a&=2+2(4)=10\end{aligned}$$

Question 5:

$$\begin{aligned}y=x^2 &\implies x=\sqrt{y}\\ \sqrt{y}(y-1)&=1\\ y(y-1)^2&=1\\ y(y^2-2y+1)&=1\\ y^3-2y^2+y-1&=0\end{aligned}$$

Question 6:

$$\begin{aligned}\alpha^3+\beta^3+\gamma^3-3\alpha\beta\gamma &= (\alpha+\beta+\gamma)(\alpha^2+\beta^2+\gamma^2-\alpha\beta-\beta\gamma-\gamma\alpha)\\ \alpha^3+\beta^3+\gamma^3 &= 3\alpha\beta\gamma\\ &= 3(-q)=-3q\end{aligned}$$

Question 7:

$$\begin{aligned}\alpha^2\beta^2+\beta^2\gamma^2+\gamma^2\alpha^2 &= (\alpha\beta+\beta\gamma+\gamma\alpha)^2-2\alpha\beta\gamma(\alpha+\beta+\gamma)\\ &= 2^2-2\left(\frac{2}{3}\right)(3)\\ &= 4-4=0\end{aligned}$$

Question 8:

$$\begin{aligned}
P(1) &= 1 - 12 + 39 - 28 = 0 \\
x^3 - 12x^2 + 39x - 28 &= (x-1)(x^2 - 11x + 28) \\
&= (x-1)(x-4)(x-7) = 0
\end{aligned}$$

Question 9:

$$\begin{aligned}
\frac{\alpha^2 + \beta^2 + \gamma^2}{\alpha\beta\gamma} &= \frac{(\alpha+\beta+\gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha)}{\alpha\beta\gamma} \\
&= \frac{b^2 - 2c}{-d} = \frac{2c - b^2}{d}
\end{aligned}$$

Easier Exam Questions

Question 1:

$$\begin{aligned}
2(\alpha+\beta) = -8 &\implies \alpha + \beta = -4 \\
\alpha\beta(-4) = 48 &\implies \alpha\beta = -12 \\
c = \alpha\beta &+ (\alpha+\beta)(-4) = -12 + 16 = 4
\end{aligned}$$

Question 2:

$$\begin{aligned}
\alpha + \beta + \gamma &= 3 \\
\alpha\beta + \alpha\gamma + \beta\gamma &= 5 \\
\alpha^2 + \beta^2 + \gamma^2 &= 3^2 - 2(5) = -1
\end{aligned}$$

Question 3:

$$\frac{\alpha+\beta+\gamma}{\alpha\beta\gamma} = \frac{5/2}{5/2} = 1$$

Question 4:

$$\begin{aligned}
\alpha^2 + \beta^2 + \gamma^2 &= \left(-\frac{2}{5}\right)^2 - 2\left(-\frac{1}{5}\right) = \frac{14}{25} \\
\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2 &= \alpha\beta\gamma(\alpha+\beta+\gamma) = \left(\frac{3}{5}\right)\left(-\frac{2}{5}\right) = -\frac{6}{25}
\end{aligned}$$

Question 5:

$$\begin{aligned}
\alpha(\beta+1) + \beta(\gamma+1) + \gamma(\alpha+1) &= (\alpha\beta + \beta\gamma + \gamma\alpha) + (\alpha+\beta+\gamma) \\
&= 7 + (-5) = 2 \\
\alpha^2 + \beta^2 + \gamma^2 &= (-5)^2 - 2(7) = 11
\end{aligned}$$

Question 6:

$$\alpha^2 + \beta^2 + \gamma^2 = 6^2 - 2(4) = 28$$

Question 7:

$$\begin{aligned}
\alpha\beta\gamma &= -\frac{a}{a} = -1 \\
(-1)\alpha\beta = -1 &\implies \alpha\beta = 1
\end{aligned}$$

Question 8:

$$\alpha^2+\beta^2+\gamma^2 = 2^2 - 2(0) = 4$$
$$\frac{\alpha^2+\beta^2+\gamma^2}{\alpha\beta\gamma} = \frac{4}{-1/2} = -8$$

Question 9:

$$\begin{aligned}(\alpha-1)(\beta-1)(\gamma-1) &= \alpha\beta\gamma - (\alpha\beta+\beta\gamma+\gamma\alpha) + (\alpha+\beta+\gamma) - 1 \\ &= \frac{1}{2} - \left(-\frac{3}{2}\right) + 2 - 1 = 3\end{aligned}$$

Question 10:

$$-\frac{m}{2} = 5 \implies m = -10$$
$$(\alpha-1)(\beta-1)(\gamma-1) = -\frac{3}{2} - \frac{1}{2} + 5 - 1 = 2$$

Harder Exam Questions

Question 1:

$$3a^2 - \beta^2 = q \implies \beta^2 = 3a^2 - q$$
$$a(a^2-\beta^2) = -r \implies a(q-2a^2) = -r$$
$$\left(-\frac{p}{3}\right)q - 2\left(-\frac{p}{3}\right)^3 = -r$$
$$-\frac{pq}{3} + \frac{2p^3}{27} = -r \implies 9pq = 27r + 2p^3$$

Question 2:

$$P(x) = x(x-2)(x+2) = x^3 - 4x$$
$$\frac{(x-2)(x+2)}{x} \geq 0 \implies x \in [-2,0) \cup [2,\infty)$$

Question 3:

$$\frac{\alpha+\beta+\gamma}{\alpha\beta\gamma} = \frac{5/2}{-1/2} = -5$$

Question 4:

$$\frac{\alpha+\beta+\gamma}{\alpha\beta\gamma} = \frac{3}{-12} = -\frac{1}{4}$$
$$P(3) = 27 - 27 + 3k + 12 = 0 \implies k = -4$$

Question 5:

$$\frac{\alpha\beta+\beta\gamma+\gamma\alpha}{\alpha\beta\gamma} = \frac{4}{-2} = -2$$

Question 6:

$$\frac{\alpha\beta+\beta\gamma+\gamma\alpha}{\alpha\beta\gamma} = \frac{4}{-2} = -2$$

Question 7:

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{-5}{-2} = 2.5$$
$$\text{Value} = 2.5 - 3 = -0.5$$

Question 8:

$$u + v + w = 0, \quad uv + vw + wu = -8/9, \quad uvw = 1/9$$
$$(u+1)(v+1)(w+1) = 1/9 - 8/9 + 0 + 1 = 2/9$$
$$\text{Value} = \frac{2/9}{1/3} = \frac{2}{3}$$

Question 9:

$$\gamma = 2, \quad \alpha + \beta = 4$$
$$a = \alpha\beta + \gamma(\alpha + \beta) = 2 + 2(4) = 10$$

Question 10:

$$\tan(A+B) = \frac{5/3}{1 - (-1/3)} = \frac{5/3}{4/3} = \frac{5}{4}$$

Question 11:

$$(1-\alpha) + (1-\beta) + (1-\gamma) = 3 - (-3) = 6$$
$$(1-\alpha)(1-\beta)(1-\gamma) = 1 - (-3) + 2 - (-0.5) = 6.5$$

Question 12:

$$\left(\frac{y}{2}\right)^3 - 5\left(\frac{y}{2}\right) + 3 = 0$$
$$\frac{y^3}{8} - \frac{5y}{2} + 3 = 0$$
$$y^3 - 20y + 24 = 0$$